

Final Project: Pick and Place Design Review

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Introduction

For this assignment students were provided with the basic requirements and recommendations for how to approach a basic Pick and Place automation using a part, in this case a cylindrical object, two conveyors and a robot with a designated gripper. The design specifications on how this sequence is carried out is left to the student. For this project, A combination of basic Station Pathing, Station Logic/Station Event Management, and Rapid code was employed to allow for the operator to easily interface with the robot and select the kind of duty they wish the robot to perform.

Design

There have been two iterations for the design of robot simulation. The first one was complicated with some unnecessary station logic and complicated unnecessary rapid code. My initial set up created many errors because of all the conditional station logic setups for the conveyor and pick and place sequence. The second iteration, I decided to lean into simplicity and familiarity. I created a similar set up to the VL11 station logic. This means two plane sensors, two linear movers, two SRLatches, a positioner, an attacher, a detacher, two digital inputs and 3 digital outputs. I used a digital output instead of the simulation start button to control when the conveyor moves. I used the Plane sensors in series with the SR latches in series with the linear movers to control when the Cylinder moves after being at the right point in the sequence. I used a plane sensor to reposition the cylinder to the original position once it reached the right point. Then writing the rapid code was very simple. I created two procedures, along with the Main. One for the Continuous execution mode, and One for a designated number of cycles. In the main, I used the TP read Function command to communicate with the operator. Here I asked the operator what kind procedure they want to run, and use a series of if statements to coordinate what procedure to call. Within the run N cycles, I used a tpreadnum to ask the operator how many cycles to run. Then I used a for loop using the variable nCycles to run the procedure and pick and place path that number of cycles. I then created a while loop for the run continuous and ran the procedure indefinitely within that while loop. Added some TP writes to communicate with the operator in between and finished the project. I ran into a lot of design flaws and errors with the previous design as I was not competent with some components. Mainly I could not get the sink and source to integrate with my existing pick and place station logic. I Also ran into a small bug which is that the flex pendant only runs the code, not the simulation. This means that the simulation events smart component did not ever go off and the simulation would not begin. I got around this by integrating a station signal to act as my “simulation start” and it was better too since it allowed me to control when the conveyor began and stopped using pulse do and wait do in my.